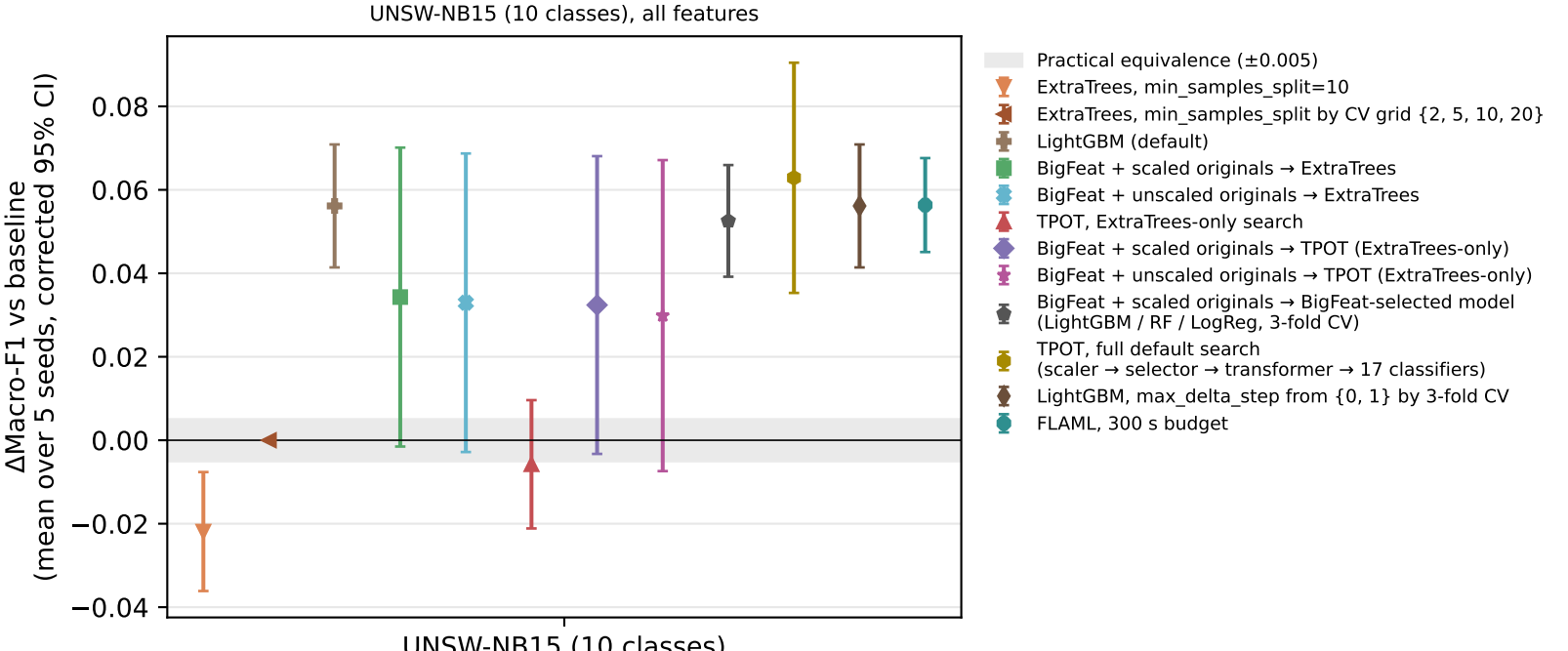
UNSW-NB15 (10 classes), all features
ΔMacro-F1 vs baseline
(mean over 5 seeds, corrected 95% CI)
0.08
0.06
0.04
0.02
0.00
−0.02
−0.04
UNSW-NB15 (10 classes)
Practical equivalence (±0.005)
ExtraTrees, min_samples_split=10
ExtraTrees, min_samples_split by CV grid {2, 5, 10, 20}
LightGBM (default)
BigFeat + scaled originals → ExtraTrees
BigFeat + unscaled originals → ExtraTrees
TPOT, ExtraTrees-only search
BigFeat + scaled originals → TPOT (ExtraTrees-only)
BigFeat + unscaled originals → TPOT (ExtraTrees-only)
BigFeat + scaled originals → BigFeat-selected model
(LightGBM / RF / LogReg, 3-fold CV)
TPOT, full default search
(scaler → selector → transformer → 17 classifiers)
LightGBM, max_delta_step from {0, 1} by 3-fold CV
FLAML, 300 s budget